Quantitative Research Methods

Advanced Module A9: Deep Learning

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The advanced modules

Module	Topic
A1	Randomised Controlled Trials — potential outcomes, randomisation, power
A2	Explainable AI with Shapley Values — axioms, exact and Monte-Carlo Shapley
A3	Conformal Prediction — split conformal, CQR, prediction sets
A4	GLMs and Splines — exponential family, overdispersion, P-splines
A5	Support Vector Machines — margins, the cost C, kernels
A6	Survival Analysis — censoring, Kaplan–Meier, Cox
A7	Multiple Testing — FWER, Bonferroni and Holm, FDR
A8	Unsupervised Learning — PCA, K-means, hierarchical clustering
A9	**Deep Learning** — MLPs, convolutions, training, regularisation

Nine optional, self-study modules that sit outside the taught chapter plan; today's module is highlighted. Each is a full deck in the course house style with a companion lab notebook.

This module assumes Chapter 3 (least squares and the linear predictor) and Chapter 5 (validation and overfitting). It was ISLP Chapter 10 and a taught session in earlier editions of this course.

Contents

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this module

Symbol	Meaning
X_j, p	input feature j; number of input features
K	number of hidden units in a layer
$g(\cdot)$	activation function (sigmoid, tanh, ReLU)
A_k	activation of hidden unit k: $A_k = g\big(w_{k0} + \sum_j w_{kj} X_j\big)$
w_{kj}, w_{k0}	weight from input j into hidden unit k; its bias
β_k, β_0	output weight of hidden unit k; output bias
softmax	output activation $e^{z_m} / \sum_{m'} e^{z_{m'}}$ for multiclass labels
W, F, P, S	conv arithmetic (appendix): input width, filter size, padding, stride
θ	all weights and biases of the network collectively
η	learning rate of (stochastic) gradient descent
ℓ	per-observation loss (squared error, cross-entropy)

Sources and references

Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python.* Springer.

Learning objectives

By the end of this chapter you will be able to:

- **Define** a single-layer and multilayer perceptron.
- **Recognise** where convolutional networks belong — the appendix develops them as optional self-study.
- **Train** a small network with stochastic gradient descent + backpropagation.
- **Reason** about when deep learning is worth it.

Roadmap of this chapter

A tour of modern deep learning

1. Single-layer networks (logistic regression in disguise).
2. Multilayer perceptrons.
3. How they are trained (SGD, backprop) and regularised.

Convolutional networks for images sit in the **appendix**: the same ideas, developed as optional self-study rather than taught here.

Outline

What is deep learning?

A renaissance of neural networks driven by GPUs, large labelled datasets, and algorithmic improvements (dropout, batch norm, attention).

Takeaway

Dominant in image, speech, and natural-language tasks. Less consistently dominant on small to medium tabular data — on the wide customer, policy and transaction tables of a bank or an insurer, gradient boosting (Chapter 8) usually matches or beats a neural network at a fraction of the cost, and is far easier to document for model validation. Deep learning earns its keep on the *unstructured* assets those firms also hold: scanned documents, call recordings, images.

Outline

Single hidden layer

$$\boxed{\hat{f}(X) = \beta_0 + \sum_{k=1}^{K} \beta_k \, g\left(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j \right) .}$$

Reading the formula symbol by symbol

- X_j: input feature j ($j = 1, \ldots, p$); g: nonlinear activation (sigmoid, tanh, ReLU).
- w_{kj}: weight from input j into hidden unit k; w_{k0}: its bias.
- K: number of hidden units (activations $A_k = g(\cdot)$); β_k: unit k's output weight, β_0 the output bias.

Takeaway

Read it inside-out: linear combination $\to$ nonlinearity g $\to$ linear combination. Logistic regression is the *no hidden layer* case: a sigmoid applied straight to $\beta_0 + \sum_j \beta_j X_j$.

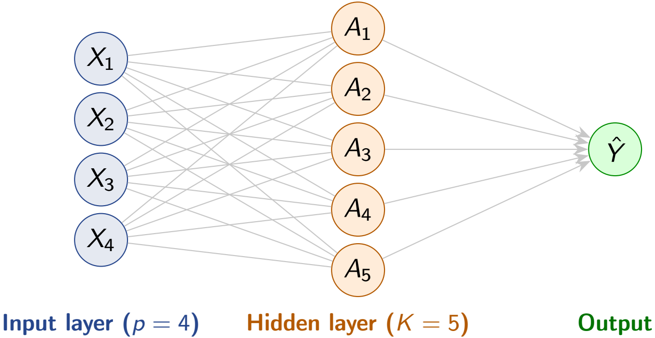
Single-layer network as a graph

Live demo — §2 of the notebook

Switch to `advanced_09_deep_learning_lab.ipynb` §2 and run the model cell: the `MLP` class builds exactly this kind of graph in code ($3 \to 32 \to 32 \to 2$, ReLU between the layers), ending in two raw logits because `nn.CrossEntropyLoss` applies the softmax itself.

Source: Figure 10.1 — ISLP, James et al. (2023).

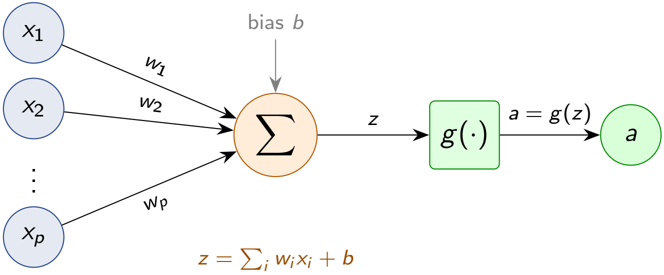
Feed-forward network as a labelled graph



Takeaway

Each hidden unit computes $A_k = g\left(w_{k0} + \sum_j w_{kj} X_j\right)$; the output linearly combines them, $\hat{Y} = \beta_0 + \sum_k \beta_k A_k$. "Fully connected" means every input feeds every hidden unit.

Anatomy of a single neuron



Takeaway

A neuron takes a weighted sum of its inputs plus a bias, then applies a nonlinearity g. Composing layers of these units builds the network above.

Activation functions: sigmoid vs. ReLU

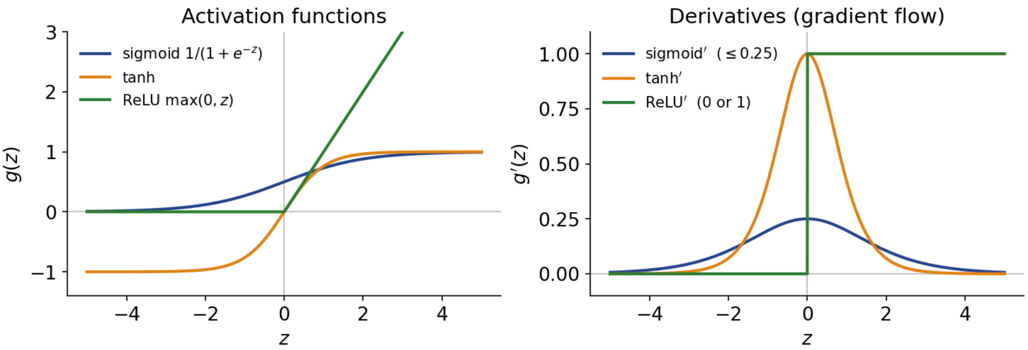
How to read this

- **Sigmoid**: $g(z) = 1/(1 + e^{-z})$, output in $(0, 1)$. Saturates at $\pm\infty$ — vanishing gradients.
- **Tanh**: $g(z) = \tanh(z) = \dfrac{e^z - e^{-z}}{e^z + e^{-z}}$, output in $(-1, 1)$; same S-shape but zero-centred.
- **ReLU**: $g(z) = \max(0, z)$. Today's default — sparser activations, faster training, better gradient flow.

Why it matters

The activation is the *only* nonlinearity in the network: remove it and stacked layers collapse to one linear map (Exercise 10.2). ReLU is preferred because its gradient stays healthy through deep stacks.

Activation functions and their gradients



Takeaway

ReLU's gradient is 1 on the active region, so signals pass undiminished; the sigmoid derivative never exceeds 0.25 and vanishes in the tails — the root of vanishing gradients in deep nets.

Exercise 10.1 — Forward pass by hand [Math]

Exercise

Consider a single hidden-layer network with input $X = (X_1, X_2) = (2, 3)$, $K = 2$ ReLU hidden units, and a linear output. The parameters are

$$\begin{aligned} \text{Unit 1: } & w_1 = (1, 1),\ w_{10} = -4, \\ \text{Unit 2: } & w_2 = (-1, 2),\ w_{20} = 0, \end{aligned} \qquad \beta_0 = 1,\ \beta_1 = 2,\ \beta_2 = -1.$$

1. Compute the two hidden activations A_1, A_2 using $g(z) = \max(0, z)$.
2. Compute the network output $\hat{f}(X)$.
3. Treating the output as a binary-classification logit, apply the sigmoid $\sigma(z) = 1/(1 + e^{-z})$ to obtain $\Pr(Y = 1 \mid X)$ and give the predicted class at threshold 0.5.

Solution — Exercise 10.1 (1/5)

Solution

Method. Work inside-out: each hidden unit first forms its affine pre-activation $z_k = w_{k0} + w_k^\top X$, then applies the nonlinearity $g(z) = \max(0, z)$ element-wise.

Solution — Exercise 10.1 (2/5)

Solution

Step 1 — pre-activations.

$$z_1 = w_{10} + w_1^\top X = -4 + (1)(2) + (1)(3) = -4 + 5 = 1,$$

$$z_2 = w_{20} + w_2^\top X = 0 + (-1)(2) + (2)(3) = -2 + 6 = 4.$$

Step 2 — ReLU activations. Both pre-activations are positive, so ReLU passes them through unchanged:

$$A_1 = \max(0, 1) = 1, \qquad A_2 = \max(0, 4) = 4.$$

Common mistake

Forgetting the bias $w_{10} = -4$ and reporting $z_1 = 5$ — always add the bias before applying g.

Solution — Exercise 10.1 (3/5)

Solution

Step 3 — output.

$$\hat{f}(X) = \beta_0 + \beta_1 A_1 + \beta_2 A_2 = 1 + (2)(1) + (-1)(4) = -1.$$

Solution — Exercise 10.1 (4/5)

Solution

Step 4 — sigmoid probability.

$$\sigma(-1) = \frac{1}{1+e^{1}} = \frac{1}{1+2.718} = 0.269.$$

Since $0.269 < 0.5$, the predicted class is $Y = 0$. *Shortcut check:* σ is monotone, so $\sigma(z) \geq 0.5 \iff z \geq 0$ — the negative logit -1 already announces class 0 before any arithmetic.

Solution — Exercise 10.1 (5/5)

Solution

Take-away. A forward pass is just alternating affine maps and element-wise nonlinearities. Note that if z_1 had been negative, ReLU would have set that unit to exactly 0, "switching it off".

Common mistake

Squashing the hidden units with the sigmoid too — here g is ReLU; only the output logit is passed through σ.

Exercise 10.2 — Why nonlinearity? [Concept]

Exercise

1. Show algebraically that a two-layer network with *identity* activations, $A^{(1)} = W^{(1)}X + b^{(1)}$ and $\hat{Y} = W^{(2)}A^{(1)} + b^{(2)}$, is equivalent to a single linear model. What does this imply for stacking layers without a nonlinearity?
2. Give two concrete advantages of ReLU over the sigmoid activation.
3. Explain the "vanishing gradient" problem of the sigmoid using its derivative.

Solution — Exercise 10.2 (1/4)

Solution

Part 1 — linear collapse. Substituting,

$$\hat{Y} = W^{(2)}\big(W^{(1)}X + b^{(1)}\big) + b^{(2)} = \underbrace{\big(W^{(2)}W^{(1)}\big)}_{W'}X + \underbrace{\big(W^{(2)}b^{(1)} + b^{(2)}\big)}_{b'} = W'X + b'.$$

This is a single affine (linear) map. Hence *any* number of linear layers collapses to one linear layer: depth buys nothing without a nonlinearity. Nonlinear g is what lets a network approximate curved decision boundaries and functions. *Common mistake: hoping the biases add flexibility — they collapse too, into the single vector b'.*

Solution — Exercise 10.2 (2/4)

Solution

Part 2 — ReLU advantages.

- *No saturation for* $z > 0$: the derivative is exactly 1, so gradients pass through undiminished, enabling deep networks to train.
- *Cheap and sparse*: $\max(0, z)$ is a single comparison; it also sets many units to exactly 0, giving sparse, efficient representations.

Solution — Exercise 10.2 (3/4)

Solution

Part 3 — vanishing gradients. For $\sigma(z) = 1/(1 + e^{-z})$ the derivative is

$$\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big) \leq 0.25,$$

with its maximum at $z = 0$, where $\sigma'(0) = 0.5 \times 0.5 = 0.25$, and $\sigma'(z) \to 0$ as $|z| \to \infty$ (already at $z = 5$, $\sigma' \approx 0.0066$). In backpropagation the layer gradients *multiply*, so through L sigmoid layers the signal shrinks by up to 0.25^L — for $L = 10$ that is $0.25^{10} \approx 10^{-6}$: it vanishes, and early layers barely learn. ReLU's derivative of 1 on the active region avoids this.

Solution — Exercise 10.2 (4/4)

Solution

Interpretation. Vanishing gradients are a property of the *product* of many small local derivatives, not of any one layer: each sigmoid layer is harmless alone, but ten in a row mute the error signal reaching the early weights.

Common mistake

Evaluating the derivative as $\sigma'(z) = z(1-z)$ — the formula plugs in the *output* $\sigma(z)$, not the pre-activation z itself.

Extended Exercise 10.1 — Tiny network by hand [Math]

Extended exercise (15 min)

A network has two inputs, one hidden layer of **two ReLU units**, and a single **sigmoid** output. Take input $x = (x_1, x_2) = (1, 2)$ and target $y = 0$, with

$$\text{Unit 1: } w_1 = (1, 1),\ b_1 = -1; \qquad \text{Unit 2: } w_2 = (0, 1),\ b_2 = 0;$$

$$\text{logit } s = v_1 a_1 + v_2 a_2 + c, \quad v = (1, -\tfrac{1}{2}),\ c = 0, \quad \hat{y} = \sigma(s).$$

Use squared-error loss $L = \frac{1}{2}(\hat{y} - y)^2$ and learning rate $\eta = 0.1$.

1. **Forward pass:** find z_k, $a_k = \max(0, z_k)$, the logit s, $\hat{y} = \sigma(s)$ and the loss L.
2. **Backprop:** write the chain rule for $\partial L / \partial w_{11}$ (the $x_1 \to$unit-1 weight) and evaluate each factor.
3. **Update:** take one gradient-descent step on w_{11} and interpret it.

Solution — Extended Exercise 10.1 (1/6)

Solution — Forward pass

Method. Alternate affine maps and nonlinearities, $z_k \to a_k \to s \to \hat{y} \to L$, and keep every intermediate value — backpropagation will reuse each one.

Solution — Extended Exercise 10.1 (2/6)

Solution — Forward pass

Hidden pre-activations and ReLU.

$$z_1 = (1)(1) + (1)(2) - 1 = 2, \qquad a_1 = \max(0, 2) = 2,$$

$$z_2 = (0)(1) + (1)(2) + 0 = 2, \qquad a_2 = \max(0, 2) = 2.$$

Output logit and sigmoid.

$$s = (1)(2) + (-\tfrac{1}{2})(2) + 0 = 1, \qquad \hat{y} = \sigma(1) = \frac{1}{1 + e^{-1}} = \frac{1}{1 + 0.368} = 0.731.$$

Squared-error loss (target $y = 0$, so the error is $\hat{y}$ itself):

$$L = \tfrac{1}{2}(\hat{y} - y)^2 = \tfrac{1}{2}(0.731)^2 = 0.267.$$

Solution — Extended Exercise 10.1 (3/6)

Solution — Backpropagation: the chain rule

The weight w_{11} reaches the loss along $z_1 \to a_1 \to s \to \hat{y} \to L$:

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial s} \frac{\partial s}{\partial a_1} \frac{\partial a_1}{\partial z_1} \frac{\partial z_1}{\partial w_{11}}.$$

Evaluate each factor:

Solution — Extended Exercise 10.1 (4/6)

Solution — Backpropagation: the chain rule

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y = 0.731, \qquad \frac{\partial \hat{y}}{\partial s} = \sigma(s)\big(1 - \sigma(s)\big) = (0.731)(0.269) = 0.197,$$

$$\frac{\partial s}{\partial a_1} = v_1 = 1, \qquad \frac{\partial a_1}{\partial z_1} = 1\{z_1 > 0\} = 1, \qquad \frac{\partial z_1}{\partial w_{11}} = x_1 = 1.$$

The first two factors combine into the reusable "error signal" $\delta = (\hat{y} - y)\,\sigma'(s) = 0.731 \times 0.197 = 0.144$.

Common mistake

Evaluating $\sigma'(s)$ by plugging $s = 1$ into $u(1-u)$ — the u in $\sigma'(s) = u(1-u)$ is $u = \sigma(s) = 0.731$.

Solution — Extended Exercise 10.1 (5/6)

Solution — Gradient step and interpretation

Multiply the factors.

$$\frac{\partial L}{\partial w_{11}} = 0.731 \times 0.197 \times 1 \times 1 \times 1 = 0.144.$$

Gradient-descent step ($\eta = 0.1$):

$$w_{11} \leftarrow w_{11} - \eta \frac{\partial L}{\partial w_{11}} = 1 - 0.1(0.144) = 0.986.$$

Check: redoing the forward pass with $w_{11} = 0.986$ gives $s = 0.986$, $\hat{y} = 0.728$, $L = 0.265 < 0.267$ — the loss indeed fell.

Solution — Extended Exercise 10.1 (6/6)

Solution — Gradient step and interpretation

Interpretation. $\hat{y} = 0.731$ overshoots the target 0, so the gradient is positive and the step *lowers* w_{11}, pulling $\hat{y}$ down. The shared factor $\delta = (\hat{y} - y)\,\sigma'(s) = 0.144$ is the output "error signal": backprop reuses it for *every* weight, multiplying only by that weight's local input. Had z_1 been negative, $\partial a_1/\partial z_1 = 0$ would kill the gradient — a "dead" ReLU unit stops learning.

Outline

Multilayer perceptron (MLP)

Stack hidden layers:

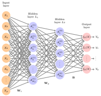
Forward pass

$$
\begin{aligned}
A^{(1)} &= g(W^{(1)}X + b^{(1)}),\\
A^{(2)} &= g(W^{(2)}A^{(1)} + b^{(2)}),\\
&\vdots\\
\hat{Y} &= h(W^{(L)}A^{(L-1)} + b^{(L)}).
\end{aligned}
$$

How to read this

h is the output activation: identity for regression, softmax for multiclass classification. A first layer can also *embed* a high-cardinality identifier (SKU, store) as a short learned vector rather than a wide one-hot block.

MLP as a graph: stacking hidden layers



Takeaway

Each layer transforms the previous layer's features; stacking them lets the network build progressively more abstract representations from the raw inputs.

History: the name honours Frank Rosenblatt's perceptron (1958); backpropagation was popularised by Rumelhart, Hinton & Williams (1986).

Source: Figure 10.4 — ISLP, James et al. (2023).

Exercise 10.3 — Counting parameters [Math]

Exercise

A fully-connected classifier has the architecture

$$\underbrace{10}_{\text{inputs}} \to 32 \to 16 \to \underbrace{3}_{\text{softmax outputs}},$$

where every hidden and output unit has a bias term.

1. State the general rule for the number of trainable parameters in a dense layer with n_{in} inputs and n_{out} units.
2. Count the parameters layer by layer.
3. Give the total.

Live demo — §4 of the notebook

After the hand count, run `advanced_09_deep_learning_lab.ipynb` §4: it prints $352 + 528 + 51 = 931$ and PyTorch's own parameter count confirms it in one line.

Solution — Exercise 10.3 (1/3)

Solution

Step 1 — the rule and why. A dense layer maps $n_{\text{in}} \to n_{\text{out}}$: every input–output pair carries one weight (an $n_{\text{in}} \times n_{\text{out}}$ matrix), and each output unit adds one bias, so

$$\#\text{params} = n_{\text{in}}\, n_{\text{out}} + n_{\text{out}}.$$

The input layer itself holds *no* parameters — it only passes the feature values forward.

Step 2 — layer by layer.

$$\begin{aligned}
\text{Layer 1 } (10 \to 32): &\quad 10 \times 32 + 32 = 320 + 32 = 352, \\
\text{Layer 2 } (32 \to 16): &\quad 32 \times 16 + 16 = 512 + 16 = 528, \\
\text{Output } (16 \to 3): &\quad 16 \times 3 + 3 = 48 + 3 = 51.
\end{aligned}$$

Solution — Exercise 10.3 (2/3)

Solution

Step 3 — total.

$$352 + 528 + 51 = \boxed{931} \text{ trainable parameters.}$$

Where the parameters live. The three blocks contribute

$$352/931 \approx 38\%, \qquad 528/931 \approx 57\%, \qquad 51/931 \approx 5\% :$$

the $32 \to 16$ block dominates because its weight *product* $n_{\text{in}} n_{\text{out}}$ is largest, while all biases together $(32 + 16 + 3 = 51)$ are almost a rounding error.

Solution — Exercise 10.3 (3/3)

Solution

Take-away. Biases add only n_{out} each but are easy to forget. Most parameters live in the widest weight matrices, which is why hidden-layer width drives model size — and why parameter counting is a quick sanity check before training.

Common mistake

Attaching parameters to the 10 input units, or dropping the $+n_{\text{out}}$ bias term — both corrupt the total.

Outline

Loss and optimisation

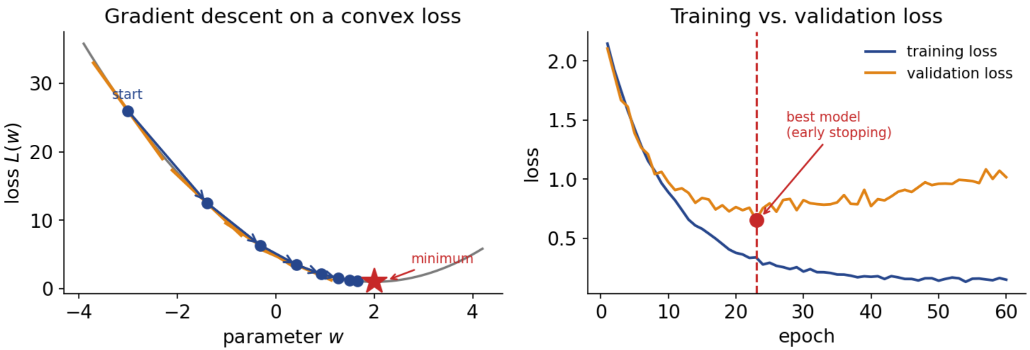
Choose a loss $L(\theta) = \sum_i \ell(y_i, \hat{y}_i; \theta)$ and minimise by stochastic gradient descent:

$$\theta \leftarrow \theta - \eta \nabla_\theta \ell_{\text{minibatch}}(\theta).$$

Reading the update

- θ: all weights and biases; $\nabla_\theta \ell$: gradient (steepest-*ascent* direction) of the loss.
- Step *against* the gradient, scaled by the learning rate η (too large diverges, too small crawls — see Exercise 10.5).
- "Stochastic": each step uses one random mini-batch (32/64/128) — a cheap, noisy estimate of the full-data gradient.
- Modern optimisers add memory: SGD+momentum, Adam, RMSprop.

Training a network: descent and early stopping



Takeaway

Gradient descent steps downhill along the loss surface (left). In practice, watch training *and* validation loss and stop at the validation minimum, before overfitting sets in (right).

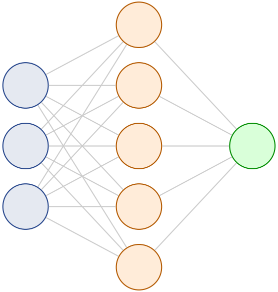
Regularisation in deep learning

Modern networks can have more parameters than data points but still generalise. Why?

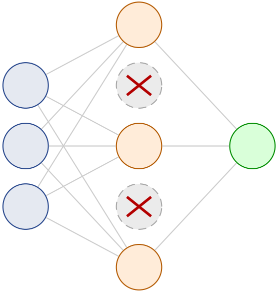
How to read this

- Implicit regularisation by SGD.
- Dropout, weight decay (ℓ_2), early stopping.
- Data augmentation.
- Architecture choice is itself a strong inductive bias.

Dropout: randomly switch off units during training



Full network



During training: 2 units dropped

Takeaway

On each mini-batch, every hidden unit is kept with probability $1 - \rho$ and otherwise *zeroed*, so the net cannot lean on any single unit and learns redundant, robust features. At test time all units are used (weights rescaled).

Exercise 10.5 — One gradient-descent step [Math]

Exercise

Consider minimising the loss $L(w) = (w-3)^2$ by gradient descent, $w \leftarrow w - \eta\, L'(w)$, starting from $w = 2$.

1. Write $L'(w)$ and evaluate it at $w = 2$.
2. Perform one update with learning rate $\eta = 0.1$.
3. Show that the update rule for the *error* $e = w - 3$ is $e \leftarrow (1-2\eta)\, e$, and deduce the range of η for which gradient descent converges.
4. What happens with $\eta = 1.5$? Interpret.

Live demo — §3 of the notebook

Every iteration of the training loop in `advanced_09_deep_learning_lab.ipynb` §3 is this update rule: run the cell and `opt.step()` applies it to all 1 250 weights at once (Adam-scaled rather than a fixed η), ending in test AUC 0.943.

Solution — Exercise 10.5 (1/4)

Solution

Part 1 — gradient. $L'(w) = 2(w - 3)$, so $L'(2) = 2(2 - 3) = -2$.

Part 2 — one step ($\eta = 0.1$).

$$w \leftarrow 2 - 0.1 \times (-2) = 2 + 0.2 = 2.2.$$

We moved from 2 toward the minimiser $w^\star = 3$: good.

Solution — Exercise 10.5 (2/4)

Solution

Part 3 — convergence range. Substituting $L'(w) = 2(w-3)$ into the update,

$$w \leftarrow w - \eta\, 2(w-3) \implies (w-3) \leftarrow (w-3) - 2\eta(w-3) = (1-2\eta)(w-3).$$

So $e_{t+1} = (1-2\eta)\, e_t$, hence $e_t = (1-2\eta)^t\, e_0$: a geometric sequence, which dies out iff its ratio satisfies $|1-2\eta| < 1$, i.e.

$$0 < \eta < 1.$$

With $\eta = 0.1$ the ratio is 0.8 — each step removes 20% of the remaining error.

Solution — Exercise 10.5 (3/4)

Solution

Part 4 — $\eta = 1.5$. Then $1-2\eta = -2$, so $|1-2\eta| = 2 > 1$: the error *doubles* each step and flips sign. Concretely $w \leftarrow 2 - 1.5(-2) = 5$; now $|5-3| = 2 > |2-3| = 1$, and iterating,

t	0	1	2	3
w_t	2	5	-1	11
$e_t = w_t - 3$	-1	2	-4	8

The steps overshoot the minimum ever more wildly: gradient descent **diverges**. This is the classic "learning rate too large" failure — η trades off speed against stability.

Solution — Exercise 10.5 (4/4)

Solution

Interpretation. The extremes are instructive: $\eta = 0.5$ makes the ratio $1 - 2\eta = 0$ and lands on $w^\star = 3$ in *one* step (possible only because this loss is exactly quadratic), while a tiny η converges but crawls.

Common mistake

Concluding "smaller is always safer, so smaller is better" — at $\eta = 0.01$ the ratio is 0.98 and hundreds of steps are needed here.

Outline

Python lab — run it live in the notebook

Companion notebook: `advanced_09_deep_learning_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- §1, *Data* — loading `Default`, the train/test split and standardisation;
- §2, *PyTorch model* — the `MLP` class, seeded with `torch.manual_seed(1)` so the AUC below is reproducible;
- §3, *Training loop* — mini-batches via `DataLoader`, then the test AUC;
- §4, *Counting the parameters* — Exercise 10.3 re-done in code and confirmed by PyTorch; the convolution count (Exercise 10.4) follows in the same cell and is optional, appendix material;
- §5, *Does the network actually beat logistic regression?* — the honest tabular comparison.

Requires `torch`; §6, *Exercises* ends the notebook with hands-on tasks.

An MLP in PyTorch

```
import torch # tensors + autograd (the gradients)
from torch import nn, optim # layer library, optimisers

class MLP(nn.Module): # a model = a subclass of nn.Module
    def __init__(self, p, k=32, K=2): # p inputs, k hidden units, K classes
        super().__init__() # register the module; must come first
        self.net = nn.Sequential( # __init__ DECLARES layers + weights
            nn.Linear(p, k), # affine map W x + b (p -> k)
            nn.ReLU(), # nonlinearity max(0, z) (Exercise 10.2)
            nn.Dropout(0.2), # zero 20% of units while training
            nn.Linear(k, k), nn.ReLU(), nn.Dropout(0.2), # 2nd hidden layer
            nn.Linear(k, K)) # K logits, no softmax (the loss adds it)
    def forward(self, x): # forward() RUNS those layers on a batch x
        return self.net(x) # model(x) calls this; autograd records each op

model = MLP(p=Xtr.shape[1]) # p predictors in, k=32 hidden units per layer
opt = optim.Adam(model.parameters(), lr=1e-3) # Adam owns the weights it moves
```

Exercise 10.6 — Softmax and cross-entropy [Python]

Exercise

A 3-class classifier outputs the logits $z = (2.0,\ 1.0,\ 0.1)$ for one observation whose true label is class 0.

1. By hand, compute the softmax probabilities $p_k = e^{z_k} / \sum_j e^{z_j}$.
2. Compute the cross-entropy loss $\ell = -\log p_y$ for the true class $y = 0$.
3. Write short NumPy code that reproduces both and prints them.

Run this live

The worked solution sits in `advanced_09_deep_learning_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Exercise 10.6 — Softmax and cross-entropy [Python]*.

Solution — Exercise 10.6 (1/5)

Solution

Method. Softmax exponentiates each logit (making it positive) and divides by the sum, turning arbitrary scores into a probability vector; cross-entropy then scores only the probability given to the *true* class.

Step 1 — exponentiate.

$$e^{2.0} = 7.389, \quad e^{1.0} = 2.718, \quad e^{0.1} = 1.105, \qquad \sum = 11.212.$$

Solution — Exercise 10.6 (2/5)

Solution

Step 2 — softmax probabilities.

$$p = \left(\frac{7.389}{11.212},\ \frac{2.718}{11.212},\ \frac{1.105}{11.212}\right) = (0.659,\ 0.242,\ 0.099).$$

Check: the three probabilities sum to 1.000.

Step 3 — cross-entropy for $y = 0$.

$$\ell = -\log p_0 = -\log(0.659) = 0.417.$$

Common mistake: using $\log_{10}$ *instead of the natural log* $(-\ln 0.659 = 0.417)$*.*

Solution — Exercise 10.6 (3/5)

Solution

```python
import numpy as np # numerics library

z = np.array([2.0, 1.0, 0.1]) # the raw scores (logits) a net emits
p = np.exp(z) / np.exp(z).sum() # softmax: make positive, rescale to sum 1
print("probs:", np.round(p, 3)) # [0.659 0.242 0.099] -- sums to 1

y = 0 # index of the TRUE class
loss = -np.log(p[y]) # cross-entropy reads off p[y] only
print("loss:", round(float(loss), 3)) # 0.417 = -log(0.659)
```

How to read this

Line 3 holds *logits* — unbounded scores, not probabilities; line 4 turns them into a probability vector, line 8 keeps only `p[0]`. PyTorch fuses both into `nn.CrossEntropyLoss`.

Solution — Exercise 10.6 (4/5)

Solution — Interpretation

The model assigns the highest probability (0.659) to the correct class, so the loss is modest. A perfect prediction ($p_0 = 1$) gives loss 0; a confidently wrong prediction sends $-\log p_0 \to \infty$ — cross-entropy punishes confident errors hardest, exactly the gradient signal a classifier needs.

Solution — Exercise 10.6 (5/5)

Solution — Interpretation

Why the "log-sum-exp" trick works. Softmax is shift-invariant: dividing top and bottom by e^c gives $p_k = e^{z_k - c} / \sum_j e^{z_j - c}$, unchanged for any c. Choosing $c = \max_k z_k = 2.0$ here,

$$e^0 = 1, \quad e^{-1} = 0.368, \quad e^{-1.9} = 0.150, \qquad p = \frac{1}{1.518}(1,\ 0.368,\ 0.150) = (0.659,\ 0.242,\ 0.09$$

the same probabilities — but every exponent is now ≤ 0, so nothing can overflow no matter how large the logits get.

Common mistake

Applying softmax yourself and then calling PyTorch's `nn.CrossEntropyLoss` — that loss already includes the softmax and expects raw logits; softmaxing twice silently distorts the loss.

Extended Exercise 10.3 — Overfitting and regularisation [Python]

Extended exercise (15 min)

Fit a **PyTorch** multilayer perceptron to the `Default` data, *watch it overfit*, then tame it. Encode `student` as 0/1, standardise `(balance, income, student)`, and predict `default`.

1. Train on a *small* subset ($n_{\text{train}} = 200$) with a wide MLP (two hidden layers of 128, no dropout) and no weight decay, recording the training and validation cross-entropy after every epoch.
2. Where does the validation loss turn upward, and what does the training loss do?
3. Refit with strong L2 regularisation (`weight_decay=0.1` in Adam); compare the validation loss and say what *early stopping* would choose.

Run this live

The worked solution sits in `advanced_09_deep_learning_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise 10.3 — Overfitting and regularisation [Python]*.

Solution — Extended Exercise 10.3 (1/4)

Solution — code: load / standardise / convert to tensors

```
import pandas as pd, torch
from torch import nn, optim
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

df = pd.read_csv("Default.csv") # 10,000 credit-card customers
df["y"] = (df["default"] == "Yes").astype(int) # response -> 0/1
df["student"] = (df["student"] == "Yes").astype(int) # a net cannot multiply "Yes"
X = df[["balance", "income", "student"]].astype(float).values # 3 predictors
Xtr, Xte, ytr, yte = train_test_split(X, df["y"].values,
    train_size=200, random_state=0, stratify=df["y"]) # only 200 rows: easy to memorise
sc = StandardScaler().fit(Xtr) # scale on TRAIN stats only (no leakage)
Xtr, Xte = sc.transform(Xtr), sc.transform(Xte) # same shift/scale applied to test
# PyTorch works on tensors: features float32, labels int64
Xtr, Xte = map(lambda a: torch.tensor(a, dtype=torch.float32), (Xtr, Xte))
ytr, yte = map(lambda a: torch.tensor(a, dtype=torch.int64), (ytr, yte))
```

Solution — Extended Exercise 10.3 (2/4)

Solution — code: train and record the loss after every epoch

```
def curve(weight_decay, epochs=400):
    torch.manual_seed(1) # same starting weights each call
    net = nn.Sequential(nn.Linear(3,128), nn.ReLU(), # wide MLP: 3 -> 128
        nn.Linear(128,128), nn.ReLU(), nn.Linear(128,2)) # -> 128 -> 2 logits
    loss_fn = nn.CrossEntropyLoss() # softmax + log-loss in one call
    opt = optim.Adam(net.parameters(), lr=1e-2, # Adam is handed the weights
                     weight_decay=weight_decay) # L2 penalty lives here
    tr, va = [], [] # one loss value per epoch, for the plot
    for _ in range(epochs): # 1 epoch = 1 full-batch step here
        opt.zero_grad() # 1. wipe .grad: PyTorch ADDS to it
        loss = loss_fn(net(Xtr), ytr) # 2. forward: predict, then score
        loss.backward(); opt.step() # 3. fill every .grad 4. move weights
        with torch.no_grad(): # scoring only -- build no graph
            tr.append(loss.item()); va.append(loss_fn(net(Xte), yte).item())
    return tr, va
tr0, va0 = curve(0.0) # no penalty -> memorises the 200 training points
tr1, va1 = curve(0.1) # strong weight decay: L2 on every weight
```

Solution — Extended Exercise 10.3 (3/4)

Solution — typical result and interpretation

Typical values (exact numbers vary with seed and version):

$$\texttt{unreg: train} \to \texttt{0.00} \quad \texttt{val-min} \approx \texttt{0.10 (epoch 16)} \quad \texttt{val-final} \approx \texttt{0.93}$$
$$\texttt{decay: train} \approx \texttt{0.19} \quad \texttt{val } \textit{lower and flat} \approx \texttt{0.18}$$

- *Unregularised:* training loss dives toward 0 (the net **memorises** the 200 points), but validation loss bottoms out within the first few dozen epochs and then **climbs** almost tenfold — textbook overfitting.
- *Weight decay:* training loss stays higher, yet validation loss is *lower and flat*: the L2 penalty trades a little fit for much better generalisation.
- *Early stopping* halts at the validation minimum — a free regulariser needing no extra penalty term.

Solution — Extended Exercise 10.3 (4/4)

Solution — typical result and interpretation

Same recipe as the companion notebook (`advanced_09_deep_learning_lab.ipynb`); there the MLP adds dropout and mini-batches via `DataLoader`.

Common mistake: judging the two nets by training loss — the unregularised net "wins" at 0.00 precisely because it memorises. Only the validation curve ranks them honestly.

Outline

Module A9 in one slide

What we accomplished

Surveyed the modern deep-learning landscape and connected it to the classical methods of the earlier chapters.

- Single- and multilayer perceptrons.
- Optimisation: SGD, momentum, Adam, backpropagation.
- Regularisation: dropout, weight decay, early stopping.
- Convolutional networks for images — in the **appendix**, as optional self-study.

Key formulas at a glance

Neuron $a = g\big(\sum_j w_j x_j + b\big)$, activation g.

Single hidden layer $f(x) = \beta_0 + \sum_{k=1}^{K} \beta_k\, g\big(w_{k0} + \sum_j w_{kj} x_j\big)$.

Activations ReLU $g(z) = \max(0, z)$; sigmoid $\sigma(z) = \dfrac{1}{1 + e^{-z}}$; softmax $\dfrac{e^{z_m}}{\sum_\ell e^{z_\ell}}$.

Loss regression $\sum_i (y_i - f(x_i))^2$; classification $-\sum_i \sum_k y_{ik} \log f_k(x_i)$.

Gradient descent $\theta \leftarrow \theta - \eta\, \nabla_\theta L(\theta)$ (learning rate η).

Dense params $\#\text{params} = n_{\text{in}}\, n_{\text{out}} + n_{\text{out}}$ (weight matrix + one bias per unit).

Conv output (appendix) $\text{size} = \lfloor (W - F + 2P)/S \rfloor + 1$ (input W, filter F, pad P, stride S).

Conv params (appendix) $\#\text{params} = (F^2 d_{\text{in}} + 1) \times (\#\text{filters})$; pooling has none.

Vocabulary checklist

Term	Meaning
Activation function	Element-wise nonlinearity (ReLU, tanh, sigmoid)
Hidden unit	One node in a hidden layer
Backpropagation	Reverse-mode automatic differentiation
SGD / momentum	Stochastic gradient + velocity term
Adam	Adaptive per-parameter learning rates
Dropout	Random unit muting during training
Weight decay	ℓ_2 penalty on weights
CNN	Conv + pool + dense layers for images (appendix)
Double descent	Test-error second dip

Decision rules of thumb

- Tabular data ($n < 10^5$, $p < 10^3$): try gradient boosting first.
- Images: pretrained CNN (ResNet, EfficientNet) and fine-tune.
- Always normalise inputs and labels.
- Track training and validation loss together.
- Save the best-validation checkpoint.

Live demo — §5 of the notebook

§5 of `advanced_09_deep_learning_lab.ipynb` runs the first rule on `Default`: logistic regression AUC 0.946 vs the MLP's 0.943 — the simpler model wins on tabular data.

Common pitfalls

Don't

1. Compare a deep network to a poorly tuned baseline.
2. Forget to shuffle training data before each epoch.
3. Use too high a learning rate (loss explodes).
4. Trust accuracy without checking calibration.
5. Skip data augmentation for image / audio tasks.
6. Re-use the validation set for many tuning rounds.

What's next

This module and **A7 — Multiple Testing** (ISLP Chapter 13, also in `Chapters/Advanced/`) are the two exam-relevant self-study modules — read A7 next if you have not already:

- The multiple-comparisons problem and the family-wise error rate.
- Bonferroni and Holm corrections.
- The false discovery rate and Benjamini-Hochberg.

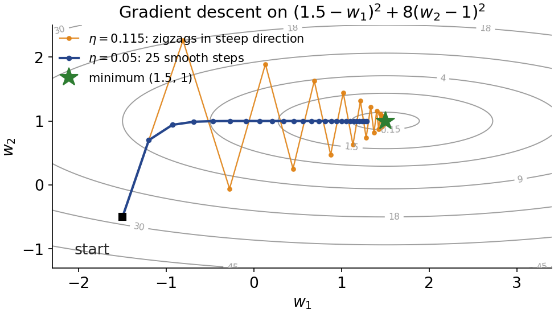
Appendix — what is in here, and why

Nothing in the appendix is needed to follow the chapter: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **Gradient descent on a loss surface: the real path** — the seeded 2-D descent figure; the learning-rate trade-off it illustrates is proved in Exercise 10.5.
- **Convolutional networks (CNNs)** — convolution, pooling, a full architecture, and Exercise 10.4 on layer arithmetic. The ideas carry over from the MLP, so the main deck spends its time on training instead; read this block when you work with images.
- **CNN architecture arithmetic (Extended Exercise 10.2)** — shapes and parameter counts through a whole network; Exercise 10.4 above does the same for a single layer.
- **Backpropagation** — the chain rule that computes the gradients. Every framework does this for you; the main deck keeps what you must know — that training is gradient descent on a loss.
- **Double descent** — why very over-parameterised models can improve again past the interpolation point. A genuinely advanced topic.

Gradient descent on a loss surface: the real path



Takeaway

Each step is perpendicular to the local contour: with $\eta = 0.05$ the loss falls from 27 to 0.05 in 25 steps — fast down the steep w_2 axis, then a crawl along the flat valley — while the too-large $\eta = 0.115$ overshoots and zigzags in the steep direction.

Why convolutions?

Image data have local, translation-invariant structure.

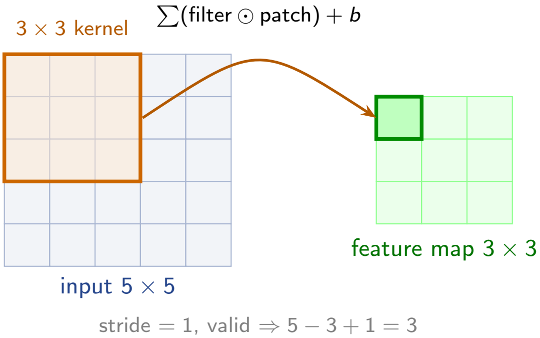
Takeaway

A cat remains a cat shifted by 10 pixels. Fully-connected networks ignore this and need much more data.

Two key layer types

- **Convolutional layers**: each unit looks at a small patch of the input; weights are shared across positions.
- **Pooling layers**: down-sample (max or average) for translation tolerance.

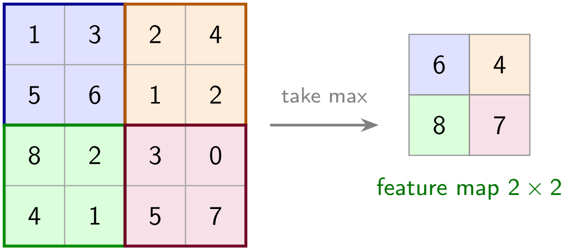
The convolution operation



Takeaway

Each output cell is a dot product of the shared-weight filter with a local input patch. Weight sharing plus locality make convolutional neural networks (CNNs) data-efficient for images.

Max-pooling: downsampling by local maxima

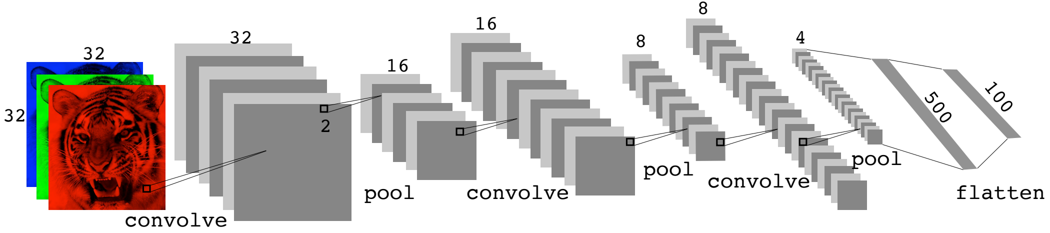


input 4×4, 2×2 windows, stride 2

Takeaway

Each outlined 2×2 window is replaced by its maximum, shrinking H, W by the stride while keeping the depth. This buys small-shift tolerance and cuts computation — and pooling has *no* trainable parameters.

A simple CNN architecture



Takeaway

The pattern repeats: convolutions extract local features, pooling shrinks the maps, and a final dense layer classifies. Early layers learn edges; deeper layers assemble them into whole objects.

Source: Figure 10.8 — ISLP, James et al. (2023).

Modern CNN tricks

- **Batch normalisation**: re-centre and scale each layer's inputs per mini-batch — stabilises and speeds up training.
- **Dropout**: randomly switch off units so the net cannot rely on any single one (detailed in the Fitting section).
- **Data augmentation**: random crops, flips, colour jitter — free extra "images" that teach invariance.
- **Skip / residual connections** (ResNet): let gradients bypass layers, enabling very deep networks.
- **Transfer learning**: reuse features pretrained on ImageNet, then fine-tune on your (smaller) task.

Takeaway

These tricks turned CNNs from finicky to reliable. Most are cheap to add and mainly improve *generalisation*.

Exercise 10.4 — CNN layer arithmetic [Math]

Exercise

A colour image of size $32 \times 32 \times 3$ is fed to a convolutional layer with 16 filters, each 5×5, stride 1, no padding ("valid"). Recall the spatial-size formula

$$O = \left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1.$$

1. Compute the output volume (height, width, depth) of the conv layer.
2. Compute the number of trainable parameters in the conv layer (include biases).
3. A 2×2 max-pooling layer (stride 2) follows. What is its output volume and how many parameters does it add?
4. Redo part (1) if instead padding $P = 2$ ("same") is used.

Solution — Exercise 10.4 (1/4)

Solution

Part 1 — conv output. With $W = 32$, $F = 5$, $P = 0$, $S = 1$:

$$O = \frac{32 - 5 + 0}{1} + 1 = 28.$$

Each filter slides over the image and produces *one* 28×28 feature map, so depth = number of filters: the output volume is $28 \times 28 \times 16$.

Solution — Exercise 10.4 (2/4)

Solution

Part 2 — conv parameters. Each filter spans the full input depth: $5 \times 5 \times 3 = 75$ weights, $+1$ bias $= 76$ per filter. With 16 filters:

$$16 \times 76 = \boxed{1216} \text{ parameters.}$$

Why so few? A dense layer from the $32 \times 32 \times 3 = 3072$ inputs to the same $28 \times 28 \times 16 = 12\,544$ outputs would need $3072 \times 12\,544 + 12\,544 \approx 38.5$ million parameters — weight sharing cuts that by a factor of about $30\,000$.

Solution — Exercise 10.4 (3/4)

Solution

Part 3 — pooling. Max-pooling with $F = 2$, $S = 2$:

$$O = \frac{28 - 2}{2} + 1 = 14,$$

giving $14 \times 14 \times 16$: pooling acts on each of the 16 channels separately, so depth is preserved. It has **no** trainable parameters — it only takes a maximum.

Solution — Exercise 10.4 (4/4)

Solution

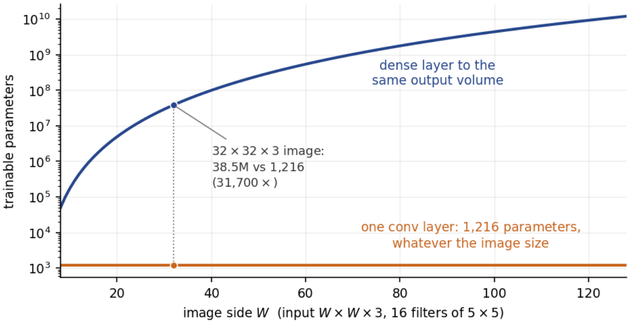
Part 4 — with padding $P = 2$.

$$O = \frac{32 - 5 + 2(2)}{1} + 1 = \frac{31}{1} + 1 = 32.$$

Output $32 \times 32 \times 16$: "same" padding preserves the spatial size. In general, stride-1 "same" padding takes $P = (F-1)/2$ — here $(5-1)/2 = 2$ — so conv layers can be stacked without shrinking the maps.

Common mistake: adding a $+2P$ term for the pooling layer (it uses $P = 0$) or forgetting to divide by the stride before adding 1.

Why convolution wins: the parameter count, drawn



Takeaway

Exercise 10.4 as a curve. The dense layer's cost grows with the *fourth power* of the image side; the convolution stays at 1 216 parameters at every size, because the same 5×5 filters slide everywhere. Weight sharing, not depth, is what makes images tractable.

Extended Exercise 10.2 — CNN architecture arithmetic [Math]

Extended exercise (15 min)

A $64 \times 64 \times 3$ RGB image passes through the stack below (convolutions use the padding shown; pooling is max-pooling, no padding). Recall $O = \lfloor (W - F + 2P)/S \rfloor + 1$.

Layer	Kernel	Stride / Pad	Filters
Conv1	3×3	1 / 1	32
Pool1	2×2	2 / 0	—
Conv2	3×3	1 / 1	64
Pool2	2×2	2 / 0	—
Dense	softmax	—	10

1. Give the output volume ($H \times W \times$ depth) after each layer.
2. Count the learnable parameters of every layer, and the total.
3. What input region does one unit of the final map, after Pool2, see (receptive field)?

Solution — Extended Exercise 10.2 (1/3)

Solution — spatial dimensions

Apply $O = (W - F + 2P)/S + 1$ to H and W; depth = number of filters (pooling keeps depth).

$$
\begin{aligned}
\text{Conv1}: (64 - 3 + 2)/1 + 1 = 64 &\quad\Rightarrow\quad 64 \times 64 \times 32, \\
\text{Pool1}: (64 - 2)/2 + 1 = 32 &\quad\Rightarrow\quad 32 \times 32 \times 32, \\
\text{Conv2}: (32 - 3 + 2)/1 + 1 = 32 &\quad\Rightarrow\quad 32 \times 32 \times 64, \\
\text{Pool2}: (32 - 2)/2 + 1 = 16 &\quad\Rightarrow\quad 16 \times 16 \times 64.
\end{aligned}
$$

Flattening the last volume gives $16 \times 16 \times 64 = 16384$ features into the dense layer. "Same" padding preserves H, W; each pool halves them.

Common mistake

Flattening only $16 \times 16 = 256$ features — the depth counts too: $16 \times 16 \times 64 = 16\,384$.

Solution — Extended Exercise 10.2 (2/3)

Solution — learnable parameters

A conv filter spans the *full input depth*, so $\# = (F^2 d_{\text{in}} + 1) \times (\#\text{filters})$; pooling has none.

$$\begin{aligned}
\text{Conv1} &: (3^2 \cdot 3 + 1) \times 32 = 28 \times 32 = 896, \\
\text{Conv2} &: (3^2 \cdot 32 + 1) \times 64 = 289 \times 64 = 18\,496, \\
\text{Dense} &: 16384 \times 10 + 10 = 163\,850, \\
\text{Pool1, Pool2} &: 0.
\end{aligned}$$

$$\text{Total} = 896 + 18\,496 + 163\,850 = \boxed{183\,242} \text{ parameters.}$$

Solution — Extended Exercise 10.2 (3/3)

Solution — receptive field and interpretation

Method. Track the receptive field r and the jump j (input pixels between neighbouring units): a layer with filter F and stride S updates $r \leftarrow r + (F-1)\,j$, then $j \leftarrow j\,S$. Start at $r=1$, $j=1$:

$$\begin{aligned}
\text{Conv1 } (F{=}3,\ S{=}1):&\quad r = 1 + 2\times 1 = 3, &\quad j = 1,\\
\text{Pool1 } (F{=}2,\ S{=}2):&\quad r = 3 + 1\times 1 = 4, &\quad j = 2,\\
\text{Conv2 } (F{=}3,\ S{=}1):&\quad r = 4 + 2\times 2 = 8, &\quad j = 2,\\
\text{Pool2 } (F{=}2,\ S{=}2):&\quad r = 8 + 1\times 2 = 10, &\quad j = 4.
\end{aligned}$$

One unit of the final map therefore sees a 10×10 patch of the original image. **Take-aways.**

(i) Convolutions are parameter-thrifty: the two conv layers use < 20k weights through *weight sharing*, while the single dense layer holds 164k — about 89% of the model. (ii) Depth plus pooling grow the receptive field, letting deep units assemble local edges into global shapes.

Backpropagation in one slide

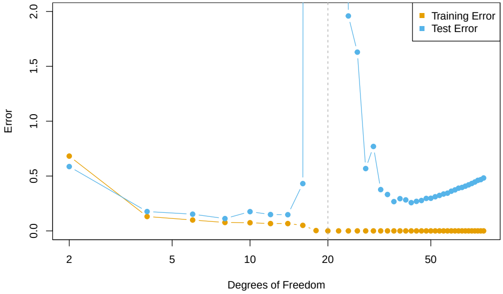
The idea in two passes

- **Forward pass**: push x through the layers, caching each activation.
- **Backward pass**: send an "error signal" δ from the loss back to every weight, multiplying local derivatives (the chain rule).
- Each weight's gradient $=$ (error signal at its output) $\times$ (its input); shared factors are reused, so the cost is $\approx$ one forward pass.

Takeaway

Backprop is just the chain rule applied from output back to input — reverse-mode automatic differentiation. Implemented in PyTorch, JAX, TensorFlow, so you rarely write it by hand.

Double descent



Why the second dip

Test error first traces the classical U-shape, then peaks at the *interpolation threshold* (the model *just* fits the data and is unstable), and falls again as extra parameters let SGD pick a smoother interpolant.

Source: Figure 10.20 — ISLP, James et al. (2023).